Year II — Mathematical Methods

Based on lectures by Various Lecturers

Notes taken by Anonymous

Michaelmas - Epiphany 2026-2027

Contents

1 Derivatives and Coordinates **2**
1.1 Vector and Scalar Functions . . . 2
1.1.1 Vector Functions . . . 2
1.1.2 Scalar Fields . . . 3
1.1.3 Vector Fields . . . 4
1.2 Inverse Functions . . . 6
1.3 Coordinate Systems . . . 7

2 Line Integrals **8**
2.1 Parametrisation and Curves . . . 8
2.2 Line Integral of Scalar Fields . . . 9
2.3 Line Integrals of Vector Fields . . . 10
2.4 Curvilinear Coordinates . . . 12

3 Differential Operators **14**
3.1 2D Surfaces . . . 14
3.2 Surface Integrals of Scalar Fields . . . 15
3.3 Surface Integrals of Vector Fields . . . 16
3.4 Volume Integrals . . . 16

Chapter 1

Derivatives and Coordinates

1.1 Vector and Scalar Functions

1.1.1 Vector Functions

Definition 1.1.1 (Vector Function)**.** A *vector function* f is a function of the form:

$$f : \mathbb{R} \to \mathbb{R}^n$$

which takes a numerical input and returns a vector

Definition 1.1.2 (Derivative of a Vector Function)**.** A vector function $\mathbf{F}(x)$ is differentiable if

$$\delta\mathbf{F} \overset{\text{def}}{=} \mathbf{F}(x+\delta x) - \mathbf{F}(x) = \mathbf{F}'(x)\delta x + o(\delta x)$$

for some $\mathbf{F}'(x)$.$\mathbf{F}'(x)$ is called the derivative of $\mathbf{F}(x)$. This is the equivalent of writing:

$$\mathbf{F}' = \frac{\mathrm{d}\mathbf{F}}{\mathrm{d}x} = \lim_{\delta x \to 0} \frac{1}{\delta x}[\mathbf{F}(x+\delta x) - \mathbf{F}(x)]$$

Example 1.1.3.

$$\mathbf{r}(t) = \begin{pmatrix} t^2 \\ e^t \\ \sin t \end{pmatrix}$$

Differentiate componentwise:

$$\mathbf{r}'(t) = \begin{pmatrix} 2t \\ e^t \\ \cos t \end{pmatrix}$$

At $t = 0$:

$$\mathbf{r}'(0) = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

Proposition 1.1.4 (Componentwise differentiation)**.** Using differential notation, the differentiability condition can be written as

$$\mathrm{d}\mathbf{F} = \mathbf{F}'(x)\mathrm{d}x$$

Given a basis $\mathbf{e}_i$ that is independent of x, vector differentiation is performed component wise:

$$\mathbf{F}'(x) = F_i'(x)\mathbf{e}_i.$$

Proposition 1.1.5 (Vector Differentiation on Operations)**.**

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{fg}) &= \frac{\mathrm{d}\mathbf{f}}{\mathrm{d}t}\mathbf{g} + \mathbf{f}\frac{\mathrm{d}\mathbf{g}}{\mathrm{d}t}\\ \frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{g}\cdot\mathbf{h}) &= \frac{\mathrm{d}\mathbf{g}}{\mathrm{d}t}\cdot\mathbf{h} + \mathbf{g}\cdot\frac{\mathrm{d}\mathbf{h}}{\mathrm{d}t}\\ \frac{\mathrm{d}}{\mathrm{d}t}(\mathbf{g}\times\mathbf{h}) &= \frac{\mathrm{d}\mathbf{g}}{\mathrm{d}t}\times\mathbf{h} + \mathbf{g}\times\frac{\mathrm{d}\mathbf{h}}{\mathrm{d}t}\end{aligned}$$

We note here that the order of operations matters for the case of the cross product since it is anti symmetric

1.1.2 Scalar Fields

Definition 1.1.6 (Scalar Field)**.** A *scalar field* f is a mapping of the form:

$$f : \mathbb{R}^n \to \mathbb{R}$$

which takes a vector and returns a numerical output.

Remark 1.1.7 (Distinction between Vector and Scalar Function)**.** A scalar function takes a position and returns a numerical output (example : potential at a point) where a vector function takes a number and maps it to a vector (example : the velocity associated to a particle at t)

Definition 1.1.8 (Limit of a Vector)**.** The limit of vectors is defined using the norm. So $\mathbf{v} \to \mathbf{c}$ iff $|\mathbf{v} - \mathbf{c}| \to 0$. So a vector approaches another vector if and only if the difference between them goes to 0. Similarly, $f(\mathbf{r}) = o(\mathbf{r})$ means $\frac{|f(\mathbf{r})|}{|\mathbf{r}|} \to 0$ as $\mathbf{r} \to \mathbf{0}$.

Definition 1.1.9 (Gradient / Derivative of a Scalar Function)**.** A scalar function $f(\mathbf{r})$ is differentiable at $\mathbf{r}$ if

$$\delta f \overset{\text{def}}{=} f(\mathbf{r}+\delta\mathbf{r}) - f(\mathbf{r}) = (\nabla f)\cdot\delta\mathbf{r} + o(\delta\mathbf{r})$$

for some vector ∇f, the gradient of f at $\mathbf{r}$.

Example 1.1.10.

$$f(x,y) = x^2 y + \sin y$$

The derivative is the gradient row/column (depending on convention). The partial derivatives are: - $\frac{\partial f}{\partial x} = 2xy$ - $\frac{\partial f}{\partial y} = x^2 + \cos y$

Gradient:

$$\nabla f(x,y) = \begin{pmatrix} 2xy \\ x^2 + \cos y \end{pmatrix}$$

At $(1,0)$:

$$\nabla f(1,0) = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

Remark 1.1.11 (Importance of Taylor Derivative Definition)**.** We see here the importance of this new way of defining a derivative in higher dimensions since dividing by a vector makes no sense if we were to try and rearrange.

Definition 1.1.12 (Directional Derivative)**.** The directional derivative of f along $\mathbf{n}$ is

$$\mathbf{n} \cdot \nabla f = \lim_{h \to 0} \frac{1}{h}[f(\mathbf{r} + h\mathbf{n}) - f(\mathbf{r})],$$

It refers to how fast f changes when we move in the direction of $\mathbf{n}$.

Remark 1.1.13 (Maximising the Directional Derivative)**.** When the derivative moves in the direction of $\mathbf{n}$ then we have the maximum possible value of the directional derivative and thus points in the direction of the greatest slope

Theorem 1.1.14 (Gradient expressed in Partial Derivative)**.** The gradient is

$$\nabla f = \frac{\partial f}{\partial x_i}\mathbf{e}_i$$

Hence we can write the condition of differentiability as

$$\delta f = \frac{\partial f}{\partial x_i}\delta x_i + o(\delta \mathbf{x}).$$

In differential notation, we write

$$\mathrm{d}f = \nabla f \cdot \mathrm{d}\mathbf{r} = \frac{\partial f}{\partial x_i}\,\mathrm{d}x_i,$$

which is the chain rule for partial derivatives.

Theorem 1.1.15 (Chain Rule for a Scalar Function)**.** Given a function $f(\mathbf{r}(u))$,

$$\frac{\mathrm{d}f}{\mathrm{d}u} = \nabla f \cdot \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}u} = \frac{\partial f}{\partial x_i}\frac{\mathrm{d}x_i}{\mathrm{d}u}.$$

Note that if we drop the $\mathrm{d}u$, we simply get

$$\mathrm{d}f = \nabla f \cdot \mathrm{d}\mathbf{r} = \frac{\partial f}{\partial x_i}\,\mathrm{d}x_i,$$

which is what we've previously had.

1.1.3 Vector Fields

Definition 1.1.16 (Vector Field)**.** A *vector field* f is a function of the form:

$$f : \mathbb{R}^n \to \mathbb{R}^m$$

which takes a vector input of dimension n and returns a vector of dimension m

Definition 1.1.17 (Derivative of a Vector Field)**.** A vector field $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable if

$$\delta\mathbf{F} \stackrel{\text{def}}{=} \mathbf{F}(\mathbf{x} + \delta\mathbf{x}) - \mathbf{F}(\mathbf{x}) = M\delta\mathbf{x} + o(\delta\mathbf{x})$$

for some $m \times n$ matrix M.M is the derivative of $\mathbf{F}$.

Theorem 1.1.18 (Derivative of a Vector field in ij notation)**.** Given an arbitrary function $\mathbf{F} : \mathbb{R}^n \to \mathbb{R}^m$ that maps $\mathbf{x} \mapsto \mathbf{y}$ and a choice of basis, we can write $\mathbf{F}$ as a set of m functions $y_j = F_j(\mathbf{x})$ such that $\mathbf{y} = (y_1, y_2, \cdots, y_m)$. Then

$$\mathrm{d}y_j = \frac{\partial F_j}{\partial x_i}\,\mathrm{d}x_i.$$

and we can write the derivative as

$$M_{ji} = \frac{\partial y_j}{\partial x_i}.$$

Note that we could have used this as the definition of the derivative. However, the original definition is superior because it does not require a selection of coordinate system.

Example 1.1.19. Take the vector field

$$F(x, y) = \begin{pmatrix} x^2 y \\ \sin(xy) \end{pmatrix}.$$

The derivative matrix is obtained by differentiating each component with respect to each variable.

$$DF(x, y) = \begin{pmatrix} \frac{\partial}{\partial x}\left(x^2 y\right) & \frac{\partial}{\partial y}\left(x^2 y\right) \\ \frac{\partial}{\partial x}(\sin(xy)) & \frac{\partial}{\partial y}(\sin(xy)) \end{pmatrix}.$$

Computing the entries: - $\frac{\partial}{\partial x}\left(x^2 y\right) = 2xy$ - $\frac{\partial}{\partial y}\left(x^2 y\right) = x^2$ - $\frac{\partial}{\partial x}(\sin(xy)) = y\cos(xy)$ - $\frac{\partial}{\partial y}(\sin(xy)) = x\cos(xy)$

So

$$DF(x, y) = \begin{pmatrix} 2xy & x^2 \\ y\cos(xy) & x\cos(xy) \end{pmatrix}$$

Example at the point $(1, 2)$:

$$DF(1, 2) = \begin{pmatrix} 4 & 1 \\ 2\cos(2) & \cos(2) \end{pmatrix}.$$

Definition 1.1.20 (Smooth Function)**.** A function is smooth if it can be differentiated any number of times. This requires that all partial derivatives exist and are totally symmetric in i, j and k (i.e.

the differential operator is commutative).

Theorem 1.1.21 (Chain Rule of Vector Fields)**.** Suppose $g : \mathbb{R}^p \to \mathbb{R}^n$ and $f : \mathbb{R}^n \to \mathbb{R}^m$. Suppose that the coordinates of the vectors in $\mathbb{R}^p, \mathbb{R}^n$ and $\mathbb{R}^m$ are u_a, x_i and y_r respectively. By the chain rule,

$$\frac{\partial y_r}{\partial u_a} = \frac{\partial y_r}{\partial x_i}\frac{\partial x_i}{\partial u_a},$$

with summation implied. Writing in matrix form,

$$M(f \circ g)_{ra} = M(f)_{ri} M(g)_{ia}.$$

Alternatively, in operator form,

$$\frac{\partial}{\partial u_a} = \frac{\partial x_i}{\partial u_a}\frac{\partial}{\partial x_i}.$$

1.2 Inverse Functions

Theorem 1.2.1 (Derivative of an Inverse Function in Vector Field)**.** Suppose $g, f : \mathbb{R}^n \to \mathbb{R}^n$ are inverse functions, i.e. $g \circ f = f \circ g = \mathrm{id(x)}$. This is where $\mathrm{id}(\mathbf{x}) = \mathbf{x}$ Suppose that $f(\mathbf{x}) = \mathbf{u}$ and $g(\mathbf{u}) = \mathbf{x}$.

Since the derivative of the identity function $\mathrm{id}(\mathbf{x})$ is the identity matrix (if you differentiate $\mathbf{x}$ wrt to $\mathbf{x}$, you get 1), we must have

$$M(f \circ g) = I.$$

Therefore we know that

$$M(g) = M(f)^{-1}.$$

We derive this result more formally by noting

$$\frac{\partial u_b}{\partial u_a} = \delta_{ab}.$$

So by the chain rule,

$$\frac{\partial u_b}{\partial x_i}\frac{\partial x_i}{\partial u_a} = \delta_{ab},$$

i.e. $M(f \circ g) = I$.

Remark 1.2.2 (Determinants)**.** For the inverse functions f and g:

$$\det M(f) \det M(g) = 1$$

1.3 Coordinate Systems

Cylindrical polars	**Spherical polars**
Conversion formulae	
$x_1 = \rho\cos\varphi$ $x_2 = \rho\sin\varphi$ $x_3 = z$	$x_1 = r\sin\theta\cos\varphi$ $x_2 = r\sin\theta\sin\varphi$ $x_3 = r\cos\theta$
Basis vectors	
$\mathbf{e}_\rho = (\cos\varphi, \sin\varphi, 0)$ $\mathbf{e}_\varphi = (-\sin\varphi, \cos\varphi, 0)$ $\mathbf{e}_z = (0, 0, 1)$	$\mathbf{e}_r = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$ $\mathbf{e}_\varphi = (-\sin\varphi, \cos\varphi, 0)$ $\mathbf{e}_\theta = (\cos\theta\cos\varphi, \cos\theta\sin\varphi, -\sin\theta)$

Chapter 2

Line Integrals

2.1 Parametrisation and Curves

Definition 2.1.1 (Curve). A curve C is a 1-d subset of $\mathbb{R}^n$, for $n > 1$.

It will be more use to us to define a curve through parametrisation.

Definition 2.1.2 (Parametrisation). *Parametrisation* is a function, $\boldsymbol{x}(t)$, which maps a real interval $[t_0, t_1]$ into $\mathbb{R}^n$. This is a vector-valued function

$$\boldsymbol{x}(t) = x_1(t)\boldsymbol{e}_1 + x_2(t)\boldsymbol{e}_2 + \ldots + x_n(t)\boldsymbol{e}_n,$$

where $\{\boldsymbol{e}_1, \boldsymbol{e}_2, \ldots, \boldsymbol{e}_n\}$ denote the Cartesian unit vectors. When $\boldsymbol{n} = \mathbf{2}$ or $\boldsymbol{n} = \mathbf{3}$, we will usually write $x_1 \equiv x, x_2 \equiv y$ and $x_3 \equiv z$.

Definition 2.1.3 (Closed and Simple Curve). If $\boldsymbol{x}(t_1) = \boldsymbol{x}(t_0)$ then the curve is called closed. A curve that does not cross itself is called simple.

Remark 2.1.4. A common parametrisation for a circle is polar coordinates. We note here however that parametrisation is not unique and a single curve may be described by multiple parametrisation.

Definition 2.1.5 (Differentaible Parametrisation). A parametrisation $\boldsymbol{x}(t)$ is differentiable if it has a well-defined tangent vector,

$$\frac{d\boldsymbol{x}}{dt} = \lim_{dt\to 0} \frac{\boldsymbol{x}(t+dt) - \boldsymbol{x}(t)}{dt},$$

so-called because its direction points along the curve at each t.

Definition 2.1.6 (Regular Parametrisation). A parametrisation is regular if $\frac{d\boldsymbol{x}}{dt} \neq 0$ everywhere, and a curve is regular if it has at least one regular parametrisation.

Definition 2.1.7 (Singular Parametrisation and Cusps). A parametrisation is singular if $\frac{d\boldsymbol{x}}{dt} = 0$ at at least one point. The singular points of a parametrisation are called susps.

Remark 2.1.8. You can construct a singular parametrisation even for a regular curve.

Remark 2.1.9. If we think of $\boldsymbol{x}(t)$ as the trajectory of a particle, then $\left|\frac{d\boldsymbol{x}}{dt}\right|$ is the speed at which the particle is moving, for each t.

Theorem 2.1.10 (Chain rule for parametrisation)**.** If $\boldsymbol{x}(t)$ and $\boldsymbol{x}(\tau)$ are two different parametrisations of the same curve, then the chain rule gives

$$\frac{d\boldsymbol{x}}{dt} = \frac{d\boldsymbol{x}}{d\tau}\frac{d\tau}{dt}.$$

2.2 Line Integral of Scalar Fields

Definition 2.2.1 (Line Integral of Scalar Field)**.** The line integral of a scalar field $f(\boldsymbol{x})$ along a curve C with parametrisation $\boldsymbol{x}(t)$, for $t \in [t_0, t_1]$, is defined as

$$\int_C f d\ell = \int_{t_0}^{t_1} f(\boldsymbol{x}(t)) \left|\frac{d\boldsymbol{x}}{dt}\right| dt$$

Proposition 2.2.2 (Formula for Scalar Line Element)**.** $\mathrm{d}\ell = \pm|\mathbf{x}'(t)|\,\mathrm{d}t$

Proposition 2.2.3 (Choice of Parametrisation does not matter)**.** The line integral of a scalar field is independent of the choice of parametrisation

Proof. If we to take two parametrisations of the same curve : $\mathbf{x}(t)$ for $t \in [t_0, t_1]$ and $\mathbf{x}(\tau)$ for $\tau \in [\tau_0, \tau_1]$, we can relate their tangent vectors by the chain rule:

$$\frac{d\boldsymbol{x}}{dt} = \frac{d\tau}{dt}\frac{d\boldsymbol{x}}{d\tau} \quad \Longrightarrow \left|\frac{d\boldsymbol{x}}{dt}\right| = \left|\frac{d\tau}{dt}\right|\left|\frac{d\boldsymbol{x}}{d\tau}\right|.$$

We now examine two cases, taking $\frac{d\mathbf{x}}{dt} > 0$ in both:

Case 1: $\mathbf{x}(\tau_0) = \mathbf{x}(t_0)$, meaning both parametrisations trace the curve in the same direction. Then:

$$\frac{d\tau}{dt} > 0 \Longrightarrow \int_{t_0}^{t_1} f(\boldsymbol{x}(t)) \left|\frac{d\boldsymbol{x}}{dt}\right| dt = \int_{t_0}^{t_1} f(\boldsymbol{x}(t)) \left|\frac{d\boldsymbol{x}}{d\tau}\right| \frac{d\tau}{dt} dt = \int_{\tau_0}^{\tau_1} f(\boldsymbol{x}(\tau)) \left|\frac{d\boldsymbol{x}}{d\tau}\right| d\tau.$$

Case 2: $\mathbf{x}(\tau_0) = \mathbf{x}(t_1)$, meaning the parametrisations trace the curve in opposite directions. Then:

$$\frac{d\tau}{dt} < 0 \quad \Longrightarrow \int_{t_0}^{t_1} f(\boldsymbol{x}(t)) \left|\frac{d\boldsymbol{x}}{dt}\right| dt = \int_{t_0}^{t_1} f(\boldsymbol{x}(t)) \left|\frac{d\boldsymbol{x}}{d\tau}\right| \left(-\frac{d\tau}{dt}\right) dt = -\int_{\tau_1}^{\tau_0} f(\boldsymbol{x}(\tau)) \left|\frac{d\boldsymbol{x}}{d\tau}\right| d\tau.$$

Once we negate the second integral, both formulas are recovered as identical.

□

Definition 2.2.4 (Rectifiable)**.** A curve which has finite length is called rectifiable. This is a weaker condition than "regular" (for example, if $\boldsymbol{x}(t)$ is Lipschitz continuous then it is rectifiable).

Remark 2.2.5 (Arclength)**.** We can derive the arc length between two points on the curve $\mathbf{r}(u)$, given by $u = u_1$ and $u = u_2$, as

$$s = \int_{u_1}^{u_2} \sqrt{\frac{d\mathbf{r}}{du} \cdot \frac{d\mathbf{r}}{du}} du \tag{2.1}$$

Proof. The tiny displacement vector is

$$d\mathbf{r} = \mathbf{r}'(u)du$$

Its magnitude gives the tiny distance travelled:

$$ds = |d\mathbf{r}|$$

Hence

$$ds = |\mathbf{r}'(u)du|$$

and since du is scalar,

$$ds = |\mathbf{r}'(u)|\,|du|$$

which becomes

$$ds = |\mathbf{r}'(u)|\,du$$

This can be directly integrated to find our arclength formula. □

2.3 Line Integrals of Vector Fields

Definition 2.3.1 (Unit Tangent Vector of Parametrisation)**.** The unit tangent vector of a parametrisation is

$$\hat{\boldsymbol{t}} = \frac{\frac{d\boldsymbol{x}}{dt}}{\left|\frac{d\boldsymbol{x}}{dt}\right|}.$$

This only depends on the direction of the parametrisation, not the "speed".

Definition 2.3.2 (Oriented Curve)**.** An oriented curve is a curve together with a specified (consistent) choice of $\hat{\mathbf{t}}$

Definition 2.3.3 (Line Integral of a Vector Field along Oriented Curve)**.** The line integral of a vector field $\boldsymbol{f}(\boldsymbol{x})$ along an oriented curve C with parametrisation $\boldsymbol{x}(t)$, for $t \in [t_0, t_1]$, is defined as the scalar line integral of the tangential component:

$$\int_C \boldsymbol{f} \cdot \hat{\boldsymbol{t}} d\ell = \int_{t_0}^{t_1} \boldsymbol{f}(\boldsymbol{x}(t)) \cdot \hat{\boldsymbol{t}} \left|\frac{d\boldsymbol{x}}{dt}\right| dt = \int_{t_0}^{t_1} \boldsymbol{f}(\boldsymbol{x}(t)) \cdot \frac{d\boldsymbol{x}}{dt} dt.$$

Example 2.3.4 (Showing Line Integral depends on Curve)**.** Take $\mathbf{F}(\mathbf{r}) = \left(xe^y, z^2, xy\right)$ and we want to find the line integral from $\mathbf{a} = (0,0,0)$ to $\mathbf{b} = (1,1,1)$.

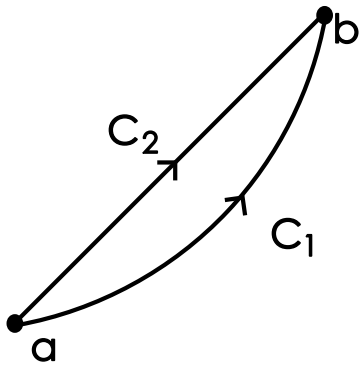


Figure 2.1: Two curves between a and b

We first integrate along the curve $C_1 : \mathbf{r}(u) = \left(u, u^2, u^3\right)$. Then $\mathbf{r}'(u) = \left(1, 2u, 3u^2\right)$, and $\mathbf{F}(\mathbf{r}(u)) = \left(ue^{u^2}, u^6, u^3\right)$. So

$$\begin{aligned}\int_{C_1} \mathbf{F} \cdot \, \mathrm{d}\mathbf{r} &= \int_0^1 \mathbf{F} \cdot \mathbf{r}'(u) \mathrm{d}u \\ &= \int_0^1 ue^{u^2} + 2u^7 + 3u^5 \, \mathrm{d}u \\ &= \frac{e}{2} - \frac{1}{2} + \frac{1}{4} + \frac{1}{2} \\ &= \frac{e}{2} + \frac{1}{4}\end{aligned}$$

Now we try to integrate along another curve $C_2 : \mathbf{r}(t) = (t, t, t)$. So $\mathbf{r}'(t) = (1, 1, 1)$.

$$\begin{aligned}\int_{C_2} \mathbf{F} \cdot \, \mathrm{d}\mathbf{r} &= \int \mathbf{F} \cdot \mathbf{r}'(t) \mathrm{d}t \\ &= \int_0^1 te^t + 2t^2 \, \mathrm{d}t \\ &= \frac{5}{3}\end{aligned}$$

We see that the line integral depends on the curve C in general, not just $\mathbf{a}, \mathbf{b}$.

Remark 2.3.5. The parametrisation must trace the curve in the direction given by $\hat{\mathbf{t}}$

Remark 2.3.6. The standard example from physics here is the work done by a particle.

Definition 2.3.7 (Circulation of $\mathbf{f}$ around C)**.** If the curve C is closed, this is often emphasised by the notation

$$\int_C \boldsymbol{f} \cdot \hat{\boldsymbol{t}} d\ell = \oint_C \boldsymbol{f} \cdot \hat{\boldsymbol{t}} d\ell$$

and the line integral is called the circulation of $\boldsymbol{f}$ around C.

Definition 2.3.8 (Piecewise Smooth Curve)**.** A piecewise smooth curve is a curve $C = C_1 + C_2 + \cdots + C_n$ with all C_i smooth with regular parametrisations. The line integral over a piecewise smooth C is

$$\int_C \mathbf{F} \cdot \, \mathrm{d}\mathbf{r} = \int_{C_1} \mathbf{F} \cdot \, \mathrm{d}\mathbf{r} + \int_{C_2} \mathbf{F} \cdot \, \mathrm{d}\mathbf{r} + \cdots + \int_{C_n} \mathbf{F} \cdot \, \mathrm{d}\mathbf{r}.$$

Example 2.3.9 (Turning two curves into piecewise smooth closed curve)**.** Take the example above, and let $C_3 = -C_2$. Then $C = C_1 + C_3$ is piecewise smooth but not smooth:

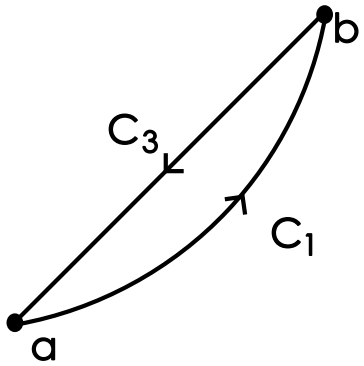


Figure 2.2: Closed line integral between a and b

Then

$$\begin{aligned}\oint_C \mathbf{F} \cdot \, \mathrm{d}\mathbf{r} &= \int_{C_1} \mathbf{F} \cdot \, \mathrm{d}\mathbf{r} + \int_{C_3} \mathbf{F} \cdot \, \mathrm{d}\mathbf{r} \\ &= \left(\frac{e}{2} + \frac{1}{4}\right) - \frac{5}{3} \\ &= -\frac{17}{12} + \frac{e}{2}\end{aligned}$$

Remark 2.3.10 (Differential form for Line Integrals)**.** You may also come across the "differential form" notation for line integrals,

$$\int_C \boldsymbol{f} \cdot \hat{\boldsymbol{t}} d\ell = \int_C \boldsymbol{f} \cdot d\boldsymbol{x}$$

2.4 Curvilinear Coordinates

Definition 2.4.1 (Curvilinear Coordinates)**.** In a curvilinear coordinate system (u, v, w), the coordinate lines - where two of the coordinates are constant and one varies - may be curved. These coordinate lines may be parametrised by the corresponding coordinate, so are given by $\boldsymbol{x}(u, v, w)$ with two of the coordinates fixed.

Definition 2.4.2 (Tangent Vectors and Unit Tangents)**.** For a system $\mathbf{x}(u, v, w)$ we have the corresponding tangent vectors:

$$\frac{\partial \boldsymbol{x}}{\partial u}, \frac{\partial \boldsymbol{x}}{\partial v}, \frac{\partial \boldsymbol{x}}{\partial w}$$

.

From these tangent vectors comes the unit tangent vectors:

$$\boldsymbol{e}_u = \frac{1}{h_u}\frac{\partial \boldsymbol{x}}{\partial u}, \quad \boldsymbol{e}_v = \frac{1}{h_v}\frac{\partial \boldsymbol{x}}{\partial v}, \quad \boldsymbol{e}_w = \frac{1}{h_w}\frac{\partial \boldsymbol{x}}{\partial w},$$

where

$$h_u = \left|\frac{\partial \boldsymbol{x}}{\partial u}\right|, \quad h_v = \left|\frac{\partial \boldsymbol{x}}{\partial v}\right|, \quad h_w = \left|\frac{\partial \boldsymbol{x}}{\partial w}\right|$$

are called the scale factors.

Remark 2.4.3. A set of tangent vectors form an orthonormal basis if the all of them are mutually orthogonal.

Remark 2.4.4. The scale factors tell us how much an infinitesimal region in (u, v, w) coordinates is stretched at different points in real space. They are crucial when calculating surface and volume integrals. We do not require systems to be orthogonal when we take their coordinates.

Definition 2.4.5 (Cylindrical and Spherical Coordinates)**.**

1. **Cylindrical coordinates** are useful for problems with axial symmetry:

$$\boldsymbol{x}(r, \theta, z) = r\cos\theta \boldsymbol{e}_1 + r\sin\theta \boldsymbol{e}_2 + z\boldsymbol{e}_3, \quad r > 0, \quad \theta \in [0, 2\pi], \quad z \in \mathbb{R},$$

   leading to scale factors $h_r = 1, h_\theta = r, h_z = 1$.

2. **Spherical coordinates** are useful for problems with spherical symmetry:

$$\boldsymbol{x}(r, \theta, \phi) = r\sin\theta\cos\phi \boldsymbol{e}_1 + r\sin\theta\sin\phi \boldsymbol{e}_2 + r\cos\theta \boldsymbol{e}_3, \quad r > 0, \quad \theta \in [0, \pi], \quad \phi \in [0, 2\pi],$$

   leading to scale factors $h_r = 1, h_\theta = r, h_\phi = r\sin\theta$.

Remark 2.4.6. It is relevant here to repeat the two coordinate basis from before:

Cylindrical polars	**Spherical polars**
Conversion formulae	
$x_1 = \rho\cos\varphi$ $x_2 = \rho\sin\varphi$ $x_3 = z$	$x_1 = r\sin\theta\cos\varphi$ $x_2 = r\sin\theta\sin\varphi$ $x_3 = r\cos\theta$
Basis vectors	
$\mathbf{e}_\rho = (\cos\varphi, \sin\varphi, 0)$ $\mathbf{e}_\varphi = (-\sin\varphi, \cos\varphi, 0)$ $\mathbf{e}_z = (0, 0, 1)$	$\mathbf{e}_r = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$ $\mathbf{e}_\varphi = (-\sin\varphi, \cos\varphi, 0)$ $\mathbf{e}_\theta = (\cos\theta\cos\varphi, \cos\theta\sin\varphi, -\sin\theta)$

Chapter 3

Differential Operators

3.1 2D Surfaces

Definition 3.1.1 (2D Surface)**.** A 2-d surface embedded in $\mathbb{R}^3$ can be defined parametrically as a map $\boldsymbol{x}(u,v)$ from $\mathbb{R}^2 \to \mathbb{R}^3$, for $u \in [u_0, u_1]$ and $v \in [v_0, v_1]$. For each value of v, the position $\boldsymbol{x}(u,v)$ describes a curve on the surface with tangent vector $\frac{\partial \boldsymbol{x}}{\partial u}$. Similarly, for each value of u, we get a curve with tangent vector $\frac{\partial \boldsymbol{x}}{\partial v}$.

Definition 3.1.2 (Unit Normal to Surface)**.** If (u, v) is a well-defined parametrisation, then their tangent vectors cannot be collinear. So the cross product of these tangent vectors allows us to define the unit normal to the surface:

$$\hat{\boldsymbol{n}} = \frac{\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}}{\left| \frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v} \right|}.$$

The sign of $\hat{\boldsymbol{n}}$ depends on the ordering of (u, v). A surface together with a specified (consistent) choice of $\hat{\boldsymbol{n}}$ is called an oriented surface.

Remark 3.1.3. Not all surfaces are orientable, meaning that a consistent choice of $\hat{\mathbf{n}}$ exists.

Example 3.1.4 (Find the outward unit normal for the torus $\boldsymbol{x}(u,v) = (R+\cos v)\cos u\boldsymbol{e}_1 + (R+\cos v)\sin u\boldsymbol{e}_2 + \sin v\boldsymbol{e}_3$, where $R > 1$ and $u, v \in [0, 2\pi)$. .]

The tangent vectors are

$$\frac{\partial \boldsymbol{x}}{\partial u} = -(R+\cos v)\sin u\boldsymbol{e}_1 + (R+\cos v)\cos u\boldsymbol{e}_2, \quad \frac{\partial \boldsymbol{x}}{\partial v} = -\sin v\cos u\boldsymbol{e}_1 - \sin v\sin u\boldsymbol{e}_2 + \cos v\boldsymbol{e}_3,$$

so

$$\begin{aligned}\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v} &= \begin{vmatrix} \boldsymbol{e}_1 & \boldsymbol{e}_2 & \boldsymbol{e}_3 \\ -(R+\cos v)\sin u & (R+\cos v)\cos u & 0 \\ -\sin v\cos u & -\sin v\sin u & \cos v \end{vmatrix} \\ &= (R+\cos v)\cos v\cos u\boldsymbol{e}_1 + (R+\cos v)\cos v\sin u\boldsymbol{e}_2 + (R+\cos v)\sin v\boldsymbol{e}_3 \\ &= (R+\cos v)\left(\cos v\cos u\boldsymbol{e}_1 + \cos v\sin u\boldsymbol{e}_2 + \sin v\boldsymbol{e}_3\right).\end{aligned}$$

This has magnitude

$$\left| \frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v} \right| = |R+\cos v|\sqrt{\cos^2 v\cos^2 u + \cos^2 v\sin^2 u + \sin^2 v} = R+\cos v,$$

where the absolute value sign can be removed since $R > 1$. So the unit normal is

$$\hat{\boldsymbol{n}} = \cos v \cos u \boldsymbol{e}_1 + \cos v \sin u \boldsymbol{e}_2 + \sin v \boldsymbol{e}_3.$$

Definition 3.1.5 (Smooth and Piecewise Smooth)**.** A surface is called smooth if $\hat{\boldsymbol{n}}$ varies continuously over the surface (e.g. a sphere). It is called piecewise smooth if it can be divided into finitely many portions each of which is smooth (e.g. a cube).

Remark 3.1.6. It is important to learn to sketch the surfaces that you are asked about

Definition 3.1.7 (Open, Closed and Boundary)**.** A surface can be either **open** if it is bounded by a curve (e.g. the cone, or a hemisphere), or **closed** if it has no boundary (e.g. the sphere or torus). The **boundary** of an open surface S is a curve (or union of curves), often written ∂S. A simple type of surface is an **explicit surface** given by the height of a two-dimensional function, $z = h(x, y)$.

3.2 Surface Integrals of Scalar Fields

Definition 3.2.1 (Surface Integral of Scalar Field)**.** If $f(\boldsymbol{x}) = f(x, y, z)$ is a scalar field and S is a surface with parametrisation $\boldsymbol{x}(u, v)$, then we define the surface integral of f over S to be

$$\int_S f dS = \int_U f(\boldsymbol{x}(u, v)) \left| \frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v} \right| du dv,$$

where U is the preimage of S in the (u, v) plane.

Remark 3.2.2. To motivate this formula, note that the area element $\left|\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}\right| du dv$ is the area on the surface in $\mathbb{R}^3$ corresponding to an infinitesimal rectangle with sides (du, dv) in the (u, v) plane.

Proposition 3.2.3. If (u, v) are (two of a set of) orthogonal curvilinear coordinates, then

$$\left| \frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v} \right| = h_u h_v.$$

Proof. Insert proof here □

Method 3.2.4 (Calculating Surface Area)**.** The surface area of S is computed by taking $f(\boldsymbol{x}) \equiv 1$, giving $|S| = \int_S dS$.

Example 3.2.5 (Use spherical coordinates to find the surface area of the sphere $x^2 + y^2 + z^2 = a^2$.)**.** The sphere radius a is given by $\boldsymbol{x}(\theta, \phi) = a \sin\theta \cos\phi \boldsymbol{e}_1 + a \sin\theta \sin\phi \boldsymbol{e}_2 + a \cos\theta \boldsymbol{e}_3$, and we know that the scale factors are $h_\theta = a$ and $h_\phi = a \sin\theta$. So from Proposition 2.1 we see that

$$\left| \frac{\partial \boldsymbol{x}}{\partial \theta} \times \frac{\partial \boldsymbol{x}}{\partial \phi} \right| = h_\theta h_\phi = a^2 \sin\theta.$$

The surface area is then

$$|S| = \int_S dS = \int_0^{2\pi} \left(\int_0^{\pi} a^2 \sin\theta d\theta \right) d\phi = a^2 \int_0^{2\pi} [-\cos\theta]_0^{\pi} d\phi = 2a^2 \int_0^{2\pi} d\phi = 4\pi a^2.$$

Proposition 3.2.6. The surface integral of a scalar field is independent of the choice of parametrisation.

Proof. insert proof here □

Remark 3.2.7. If $S \subset \mathbb{R}^2$ is a flat surface in (say) the (x, y)-plane, then $\hat{\boldsymbol{n}}$ only has a z-component, so

$$\left|\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}\right| = \left|\frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}\right| = |J|,$$

which is just the Jacobian for a change of coordinates from (x, y) to (u, v) in the plane.

3.3 Surface Integrals of Vector Fields

Definition 3.3.1 (Flux)**.** If $\boldsymbol{f} : \mathbb{R}^3 \to \mathbb{R}^3$ is a vector field and S is an oriented surface with unit normal $\hat{\boldsymbol{n}}$, we define the surface integral (or flux) of $\boldsymbol{f}$ through S as

$$\int_S \boldsymbol{f} \cdot \hat{\boldsymbol{n}} dS.$$

Again, we evaluate the integral using a specific parametrisation $\boldsymbol{x}(u, v)$, so

$$\begin{aligned}\int_S \boldsymbol{f} \cdot \hat{\boldsymbol{n}} dS &= \int_U \boldsymbol{f}(\boldsymbol{x}(u,v)) \cdot \hat{\boldsymbol{n}} \left|\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}\right| du dv \\ &= \int_U \boldsymbol{f}(\boldsymbol{x}(u,v)) \cdot \left(\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}\right) du dv\end{aligned}$$

where U is the preimage of S in the (u, v) plane.

Remark 3.3.2. A non-zero flux requires a component of $\boldsymbol{f}$ pointing "through" the surface. For given $|\boldsymbol{f}|$, the flux is largest if $\boldsymbol{f}$ is exactly perpendicular to the surface, but the flux also depends on the magnitude $|\boldsymbol{f}|$.

Definition 3.3.3 (Vector Area Element)**.** Sometimes you will see the shorthand notation $d\boldsymbol{S} = \hat{\boldsymbol{n}} dS$ for the "vector area element".

Remark 3.3.4. Unlike a scalar surface integral, a flux integral depends on the sign of $\hat{\boldsymbol{n}}$, i.e. on the orientation of the surface. You need to check that your chosen parametrisation $\boldsymbol{x}(u, v)$ gives the correct sign of $\hat{\boldsymbol{n}} = \frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}$. If not, interchange u and v.

Example 3.3.5 (Example 2.8: Calculate the outward flux of $\boldsymbol{f}(\boldsymbol{x}) = \cos^7 \boldsymbol{\theta} \boldsymbol{e}_r$ [in spherical coordinates]**.** through the hemisphere $r = a, z \geq 0$.]

Parametrising the sphere in spherical coordinates as $\boldsymbol{x}(\theta, \phi)$, recall that $\hat{\boldsymbol{n}} = \boldsymbol{e_r}$ and

$$\left|\frac{\partial \boldsymbol{x}}{\partial \theta} \times \frac{\partial \boldsymbol{x}}{\partial \phi}\right| = h_\theta h_\phi = a^2 \sin\theta.$$

Also note that the hemisphere is given by the half-range $\theta \in [0, \pi/2]$, with $\phi \in [0, 2\pi]$ as usual. So

$$\begin{aligned}\int_S \boldsymbol{f}\cdot\hat{\boldsymbol{n}}dS &= \int_0^{2\pi}\left(\int_0^{\pi/2}\boldsymbol{f}\cdot\hat{\boldsymbol{n}}\left|\frac{\partial\boldsymbol{x}}{\partial\theta}\times\frac{\partial\boldsymbol{x}}{\partial\phi}\right|d\theta\right)d\phi\\ &= \int_0^{2\pi}\left(\int_0^{\pi/2}\left(\cos^7\theta\boldsymbol{e}_r\right)\cdot\left(\boldsymbol{e}_r\right)a^2\sin\theta d\theta\right)d\phi\\ &= a^2\int_0^{2\pi}\left(\int_0^{\pi/2}\cos^7\theta\sin\theta d\theta\right)d\phi\\ &= a^2\int_0^{2\pi}\left[-\frac{1}{8}\cos^8\theta\right]_0^{\pi/2}d\phi = \frac{a^2}{8}\int_0^{2\pi}d\phi = \frac{\pi a^2}{4}.\end{aligned}$$

Definition 3.3.6 (Closed Surface Flux)**.** If the surface S is closed, this is often emphasised by writing

$$\int_S \boldsymbol{f}\cdot\hat{\boldsymbol{n}}dS = \oint_S \boldsymbol{f}\cdot\hat{\boldsymbol{n}}dS.$$

Definition 3.3.7 (Vector Valued Surface Integrals)**.** We will also meet vector-valued surface integrals of the form

$$\int_S \boldsymbol{f}dS$$

for some vector field $\boldsymbol{f}:\mathbb{R}^3\to\mathbb{R}^3$.

3.4 Volume Integrals

Definition 3.4.1 (Volume Integral)**.** The volume integral of a scalar field $f:\mathbb{R}^3\to\mathbb{R}$ over a 3 -d subregion $V\subset\mathbb{R}^3$ is

$$\int_V f dV = \int_U f(\boldsymbol{x}(u,v,w))\left|\frac{\partial\boldsymbol{x}}{\partial u}\cdot\left(\frac{\partial\boldsymbol{x}}{\partial v}\times\frac{\partial\boldsymbol{x}}{\partial w}\right)\right|dudvdw,$$

where U is the preimage of V in (u,v,w) space, for some well-defined coordinates (u,v,w).

Remark 3.4.2. The factor $\left|\frac{\partial\boldsymbol{x}}{\partial u}\cdot\left(\frac{\partial\boldsymbol{x}}{\partial v}\times\frac{\partial\boldsymbol{x}}{\partial w}\right)\right|dudvdw$ is the volume of a parallelepiped in "real space" corresponding to an infinitesimal cuboid (du,dv,dw) in (u,v,w) space.

Remark 3.4.3. If (u,v,w) are orthogonal curvilinear coordinates, then

$$\left|\frac{\partial\boldsymbol{x}}{\partial u}\cdot\left(\frac{\partial\boldsymbol{x}}{\partial v}\times\frac{\partial\boldsymbol{x}}{\partial w}\right)\right| = h_u h_v h_w.$$

Proof. insert proof here □

Remark 3.4.4. The factor $\left|\frac{\partial\boldsymbol{x}}{\partial u}\cdot\left(\frac{\partial\boldsymbol{x}}{\partial v}\times\frac{\partial\boldsymbol{x}}{\partial w}\right)\right|$ is often called the Jacobian (or Jacobian determinant). To see why, note that the scalar triple product may be written

$$\frac{\partial\boldsymbol{x}}{\partial u}\cdot\left(\frac{\partial\boldsymbol{x}}{\partial v}\times\frac{\partial\boldsymbol{x}}{\partial w}\right) = \begin{vmatrix}\frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w}\\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w}\\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w}\end{vmatrix}.$$

Compare this to the 2-d case of a flat surface in the (x,y) plane, where we had

$$\left|\frac{\partial \boldsymbol{x}}{\partial u} \times \frac{\partial \boldsymbol{x}}{\partial v}\right| = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}.$$

Method 3.4.5 (Calculating Volume)**.** The volume of V is computed by taking $f(\boldsymbol{x}) \equiv 1$, giving $|V| = \int_V dV$.

Example 3.4.6 (Let V be the region bounded by the planes $x = 0, y = 0, z = 2$ and the surface $z = x^2 + y^2$ for $x, y \geq 0$. Calculate $\int_V x dV$.)**.** First we make a sketch to establish the shape of the region. Note that the surface $z = x^2 + y^2$ is a paraboloid.

Insert Sketch

Clearly this region has cylindrical symmetry, so the integral will be easiest to evaluate in cylindrical coordinates, $\boldsymbol{x}(r, \theta, z) = r\cos\theta \boldsymbol{e}_1 + r\sin\theta \boldsymbol{e}_2 + z\boldsymbol{e}_3$.

By Proposition 2.3 the Jacobian is $\left|\frac{\partial \boldsymbol{x}}{\partial r} \cdot \left(\frac{\partial \boldsymbol{x}}{\partial \theta} \times \frac{\partial \boldsymbol{x}}{\partial z}\right)\right| = h_r h_\theta h_z = r$. In our case, the boundaries $x = 0$ and $y = 0$ correspond to $\theta = 0$ and $\theta = \pi/2$, respectively. The range in r depends on the height, with maximum given by $z = x^2 + y^2 = r^2$, so that $r(z) \in [0, \sqrt{z}]$. Therefore we have

$$\begin{aligned}
\int_V x dV &= \int_0^2 \left(\int_0^{\pi/2} \left(\int_0^{\sqrt{z}} x r dr \right) d\theta \right) dz \\
&= \int_0^2 \left(\int_0^{\pi/2} \left(\int_0^{\sqrt{z}} r^2 \cos\theta dr \right) d\theta \right) dz \\
&= \int_0^2 \left(\int_0^{\pi/2} \cos\theta \left[\frac{r^3}{3} \right]_0^{\sqrt{z}} d\theta \right) dz \\
&= \int_0^2 \frac{z^{3/2}}{3} \left(\int_0^{\pi/2} \cos\theta d\theta \right) dz \\
&= \int_0^2 \frac{z^{3/2}}{3} [\sin\theta]_0^{\pi/2} dz = \int_0^2 \frac{z^{3/2}}{3} dz = \frac{2}{15} \left[z^{5/2} \right]_0^2 = \frac{2}{15} 2^{5/2} = \frac{8\sqrt{2}}{15}.
\end{aligned}$$

Proposition 3.4.7. The volume integral of a scalar field is independent of the choice of coordinates.

Remark 3.4.8. Occasionally you will come across a vector-valued volume integral, $\int_V f dV$. An example is the centre of mass of a solid body,

$$\overline{\boldsymbol{x}} = \frac{\int_V \rho(\boldsymbol{x}) \boldsymbol{x} dV}{\int_V \rho(\boldsymbol{x}) dV}$$

where $\rho(\boldsymbol{x})$ is the distribution of mass density within the body.